

Lab Title:- STEGANOGRAPHY AND HIDDEN DATA DETECTION

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Course Name:- COMPUTER AND DIGITAL FORENSICS

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Executive summary

The lab covers a practical hands-on activity about steganography by embedding file into another file and then extracting that same file.

Lab Objectives

The objectives of this lab is to understand steganography and how important is steganography in digital forensic.

Tools and Resources Used

- Kali linux:- A penetration testing platform used for security assessments.

Kali linux tools

- steghide:- A steganography program that is able to hide data in various kinds of image- and audio-files
 - stegcracker:- A steganography brute-force utility to uncover hidden data inside files.
 - stegseek:- used together with a wordlist to search for passphrase.
 - md5sum:- Print or check MD5 (128-bit) checksums.
 - sha256sum:- Print or check SHA256 (256-bit) checksums.
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Methology

- I launched the kali linux terminal
- I used the wget command to download the image file
- I obtained the md5/sha256 hash of the image file using the md5sum/sha256sum command
- I created a text file to be embedded into the image file
- I obtained the md5/sha256 hash of the text file created using the md5sum/sha256sum command

- I embedded the text file into the image file
- I obtained the md5/sha256 hash of the image file using the md5sum/sha256sum command after embedding
- I extract the text file from the image file
- I obtained the md5/sha256 hash of the text file using the md5sum/sha256sum command after extracting

Screenshot and Evidence

SCREENSHOT SHOWING MODULE 8 REVIEW

ICDFA / COURSES
Introduction to Steganography and Hidden Data Lesson...
SBT-DF202 — Computer and Digital Forensics

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31/31 lessons 100%

MODULE 8 • LESSON 28

Introduction to Steganography and Hidden Data Lesson type

Text lesson 25 minutes Completed 02 Sep 2026

Steganography is a technique used to hide information inside another file. The carrier file may appear normal to a user, while additional information is concealed within it.

Unlike ordinary encryption, where the existence of protected information is visible, steganography attempts to conceal the presence of the hidden information itself.

What is Steganography?

The course material defines steganography as hiding information inside another file.

A typical steganography scenario may involve:

COURSE CONTENT
Lesson outline 31

- Text - 35 min
- MODULE 8 - STEGANOGRAPHY AND HIDDEN DATA ANALYSIS
 - Introduction to Steganography and Hidden Data Lesson type (Text - 25 min)
 - Least Significant Bit Substitution and Bitmap Structures (Text - 25 min)
 - Embedding and Extracting Hidden Data with Steghide (Text - 30 min)
 - Steganography Password Recovery with Wordlists and StegSeek (Text - 30 min)

Portal Guide

SCREENSHOT SHOWING PREPARATION OF THE LAB ENVIRONMENT & ORIGINAL CARRIER IMAGE AND ITS HASH

```
(kali@kali)-[~]
└─$ cd STSNOGRAPHY
(kali@kali)-[~/STSNOGRAPHY]
└─$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/steganography/_tower_original_image_for_lab.bmp
(kali@kali)-[~/STSNOGRAPHY]
└─$ ls
_tower_original_image_for_lab.bmp
(kali@kali)-[~/STSNOGRAPHY]
└─$ display _tower_original_image_for_lab.bmp
(kali@kali)-[~/STSNOGRAPHY]
└─$ md5sum _tower_original_image_for_lab.bmp
7f77e02253d5ae6426c4d0c932325ae3 _tower_original_image_for_lab.bmp
(kali@kali)-[~/STSNOGRAPHY]
└─$ imp_stat _tower_original_image_for_lab.bmp
IMAGE FILE INFORMATION
-----
Image Type: raw
Size in bytes: 9277494
Sector size: 512
(kali@kali)-[~/STSNOGRAPHY]
└─$ sha256sum
^C
(kali@kali)-[~/STSNOGRAPHY]
└─$ sha256sum _tower_original_image_for_lab.bmp
6508bebb94bd8ec0833b9f85580cda4261d89632ddc89e7f9f7093b5e653c4db _tower_original_image_for_lab.bmp
(kali@kali)-[~/STSNOGRAPHY]
└─$
```

SCREENSHOT SHOWING SECRET EVIDENCE FILE WITH ITS HASH VALUE

```
(kali@kali)-[~/STSNOGRAPHY]
└─$ nano secret.txt
(kali@kali)-[~/STSNOGRAPHY]
└─$ sha256sum secret.txt
b85e6ffb473fad2c374b08c8b5b6108574b67f0ac124cd4068a0a517486d8b85 secret.txt
(kali@kali)-[~/STSNOGRAPHY]
└─$
```

SCREENSHOT SHOWING SUCCESSFUL EMBEDDING & EXTRACTION OF EVIDENCE FILE FROM THE CARRIER IMAGE

```
(kali@kali)-[~/STSNOGRAPHY]
└─$ steghide embed -ef secret.txt -cf _tower_original_image_for_lab.bmp -p 1234
embedding "secret.txt" in "_tower_original_image_for_lab.bmp"... done
(kali@kali)-[~/STSNOGRAPHY]
└─$ md5sum _tower_original_image_for_lab.bmp
ff986330f628d36202de9cd1e1c4241b _tower_original_image_for_lab.bmp
(kali@kali)-[~/STSNOGRAPHY]
└─$ sha256sum _tower_original_image_for_lab.bmp
e727e55c1aaf82d9935e9642b24716de7774685e9ce707fe72e8d553256af60c _tower_original_image_for_lab.bmp
(kali@kali)-[~/STSNOGRAPHY]
└─$ steghide extract -sf _tower_original_image_for_lab.bmp -xf extracted_secret.txt -p 1234
wrote extracted data to "extracted_secret.txt".
```

SCREENSHOT SHOWING CONTROLLED PASSWORD DICTIONARY TEST

```
(kali㉿kali)-[~/STENOGRAPHY]
$ stegseek _tower_original_image_for_lab.bmp wordlist.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek
Secret Size: 12
[i] Found passphrase: "1234"
[i] Original filename: "secret.txt".
[i] Extracting to "_tower_original_image_for_lab.bmp.out".
Cannot determine file system type

(kali㉿kali)-[~/STENOGRAPHY]
$
```

SCREENSHOT SHOWING STUDENT-CREATED INDEPENDENT STEGANOGRAPHY EVIDENCE FILE TASK

```
(kali㉿kali)-[~/STENOGRAPHY]
$ nano MUNTAKA_BADAMASI_hidden_note.txt

(kali㉿kali)-[~/STENOGRAPHY]
$ cat MUNTAKA_BADAMASI_hidden_note.txt
STENOGRAPHY can simply be define as the process of hiding information inside another file such as image.
the following are the reason why steganography matters in digital forensics is that with steganography
1) it helps in verifying whether hidden content exist in a file
2) it helps in knowing how the information was embedded
3) it helps in knowing if the concealed information can be extracted successfully from the file it is embedded to
4) it helps in knowing which carrier file contains the concealed information.

(kali㉿kali)-[~/STENOGRAPHY]
$ sha256sum MUNTAKA_BADAMASI_hidden_note.txt
6fe0d875d75fb94df58565321ed8d5647a0f805cfb57f5aae0806dce0b2a003e MUNTAKA_BADAMASI_hidden_note.txt

(kali㉿kali)-[~/STENOGRAPHY]
$ sha256sum _tower_original_image_for_lab.bmp
2adea23551a86865abbd5d7f562ba142167e10181f732ccf3bf2aa6016e84013 _tower_original_image_for_lab.bmp

(kali㉿kali)-[~/STENOGRAPHY]
$ steghide embed -ef MUNTAKA_BADAMASI_hidden_note.txt -cf _tower_original_image_for_lab.bmp -p 2558
embedding "MUNTAKA_BADAMASI_hidden_note.txt" in "_tower_original_image_for_lab.bmp"... done

(kali㉿kali)-[~/STENOGRAPHY]
$ steghide extract -sf _tower_original_image_for_lab.bmp -xf EXTRACTED-MUNTAKA_BADAMASI_hidden_note.txt -p 2558
wrote extracted data to "EXTRACTED-MUNTAKA_BADAMASI_hidden_note.txt".

(kali㉿kali)-[~/STENOGRAPHY]
$ sha256sum EXTRACTED-MUNTAKA_BADAMASI_hidden_note.txt
6fe0d875d75fb94df58565321ed8d5647a0f805cfb57f5aae0806dce0b2a003e EXTRACTED-MUNTAKA_BADAMASI_hidden_note.txt

(kali㉿kali)-[~/STENOGRAPHY]
$ nano wordlist2.txt

(kali㉿kali)-[~/STENOGRAPHY]
$ stegseek _tower_original_image_for_lab.bmp wordlist2.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: "2558"
[i] Original filename: "MUNTAKA_BADAMASI_hidden_note.txt".
[i] Extracting to "_tower_original_image_for_lab.bmp.out".
the file "_tower_original_image_for_lab.bmp.out" does already exist. overwrite ? (y/n)
y

(kali㉿kali)-[~/STENOGRAPHY]
$
```

Conclusion

In conclusion the lab has laid a foundation and also expose me to understanding steganography and also how important is steganography in digital forensic.

Reference

- ICDFA course Material (BVDF201(computer and digital forensics)- by COMMANDANT AMINU IDRIS
 - ICDFA Recorded sessions
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